

Hybrid Optimal and Residual Reinforcement Learning Controller for 6-DOF Ball-and- Plate Stabilization

Team Members:

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College of Engineering

Introduction

Marble balancing/ pos tracking

Optimal Control LQR:

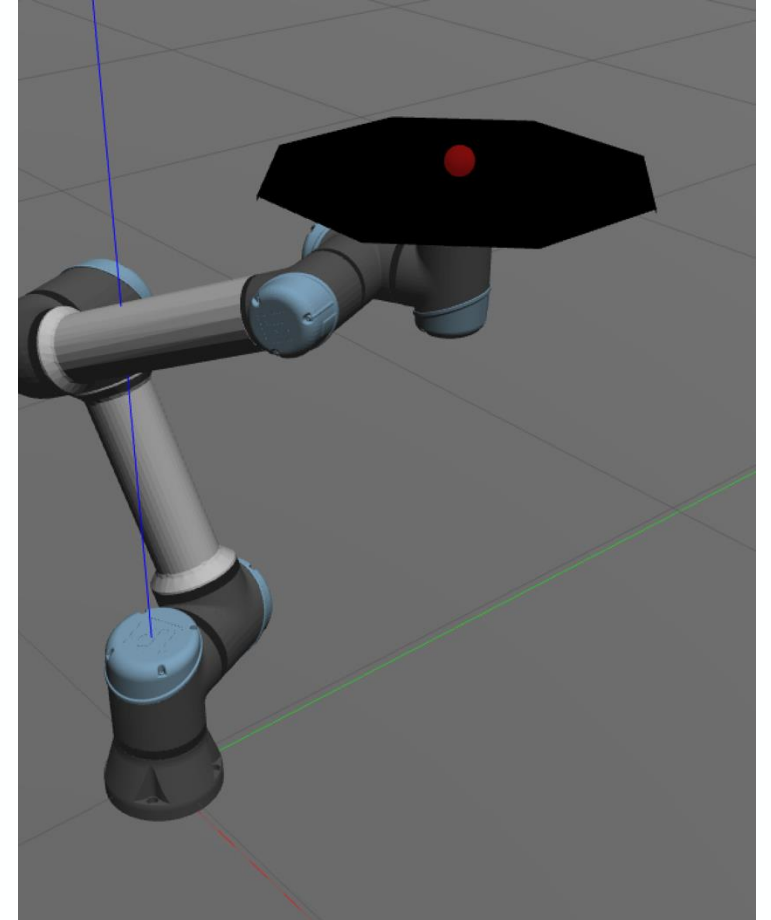
- Model-based
- Linearized
- Optimal / Robust
- Unconsidered Saturations

Reinforcement Learning:

- Model-Free / Data-Driven
- Can learn NonLin Behavior
- Adaptable

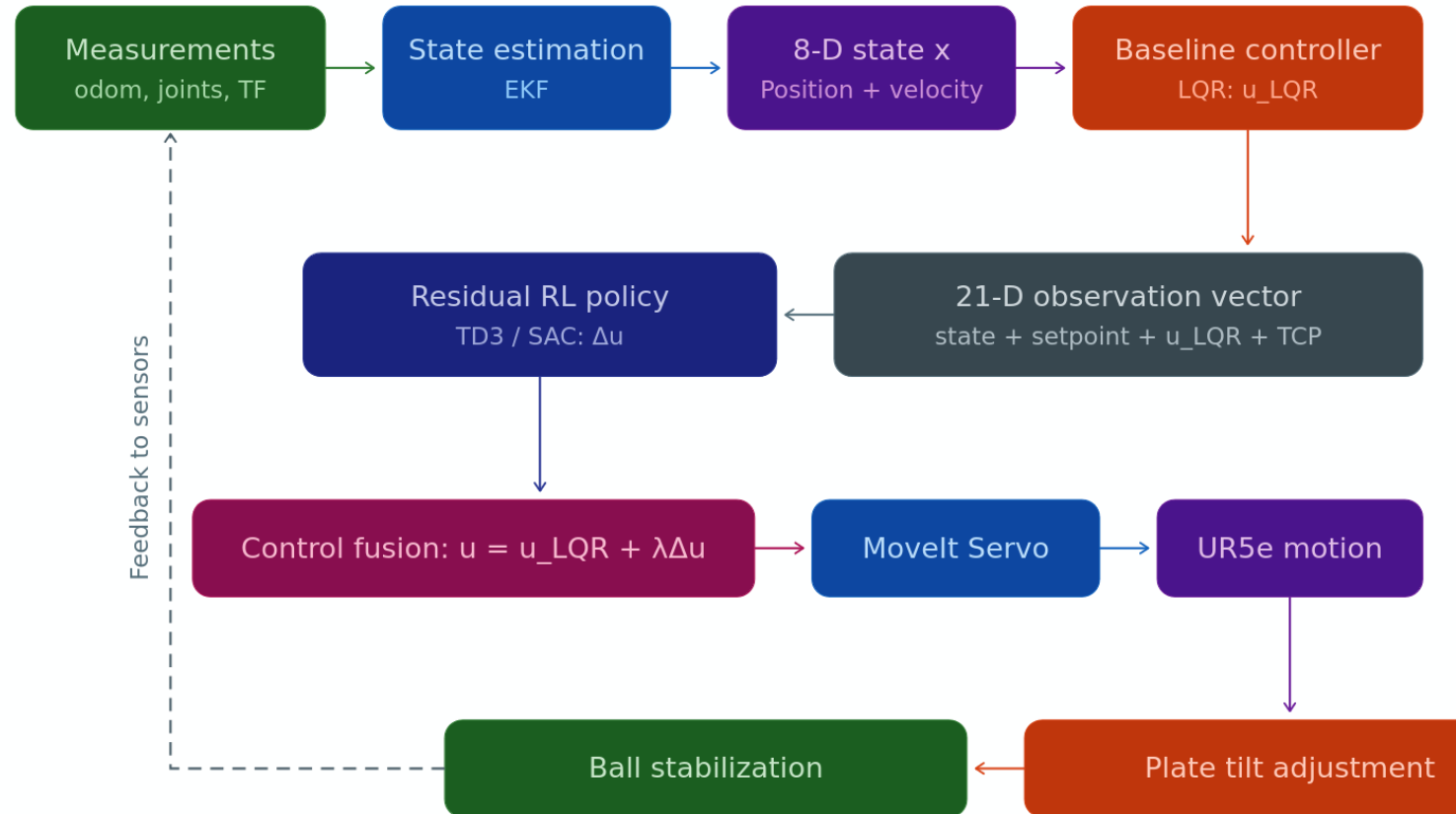


**Hybrid Optimal and Residual Reinforcement
Learning Controller**



Methods

System Architecture



Methods

Optimal LQR Controller

$$\ddot{x} = -C \alpha, \quad \ddot{y} = -C \beta,$$

$$C = \frac{M_b g}{M_b + I_b/r^2} = \frac{5}{7} g \approx 7.0 \text{ m/s}^2,$$

$$\dot{\omega} = \frac{\omega_{\text{cmd}} - \omega}{T}.$$



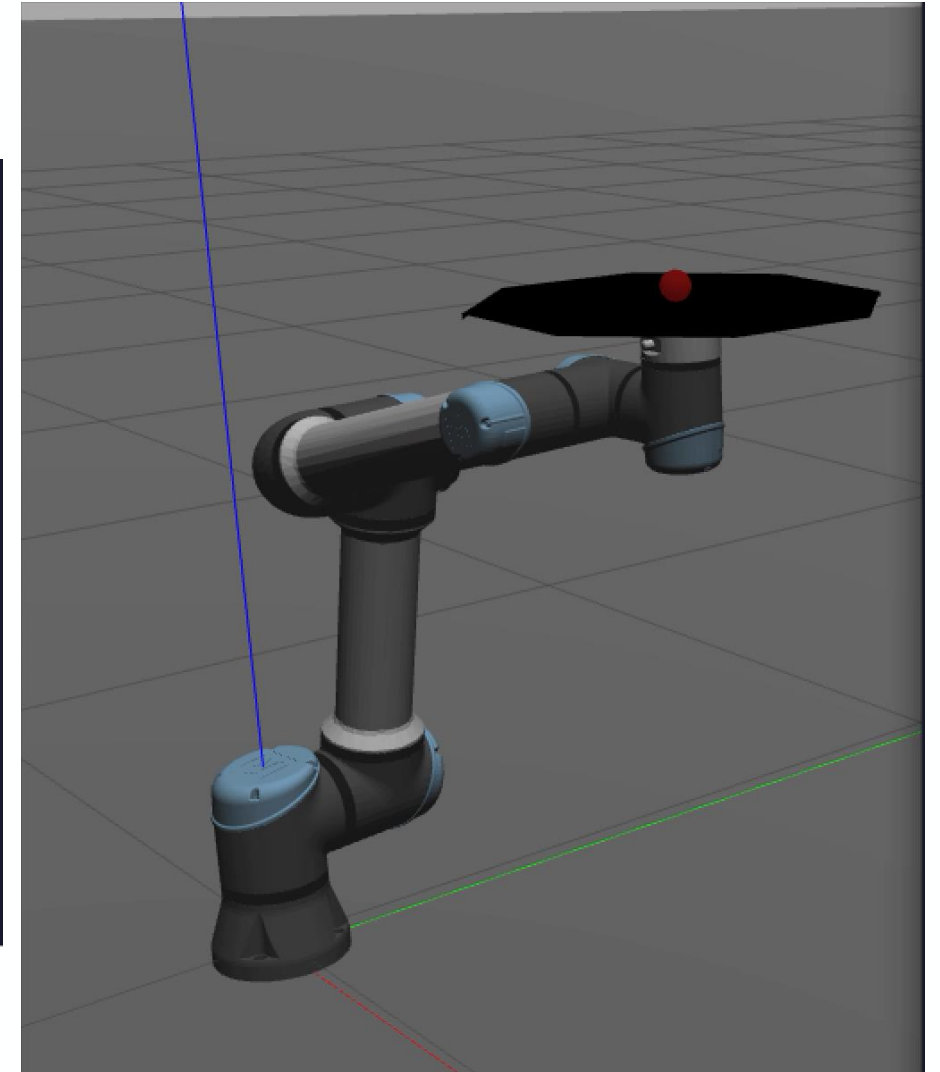
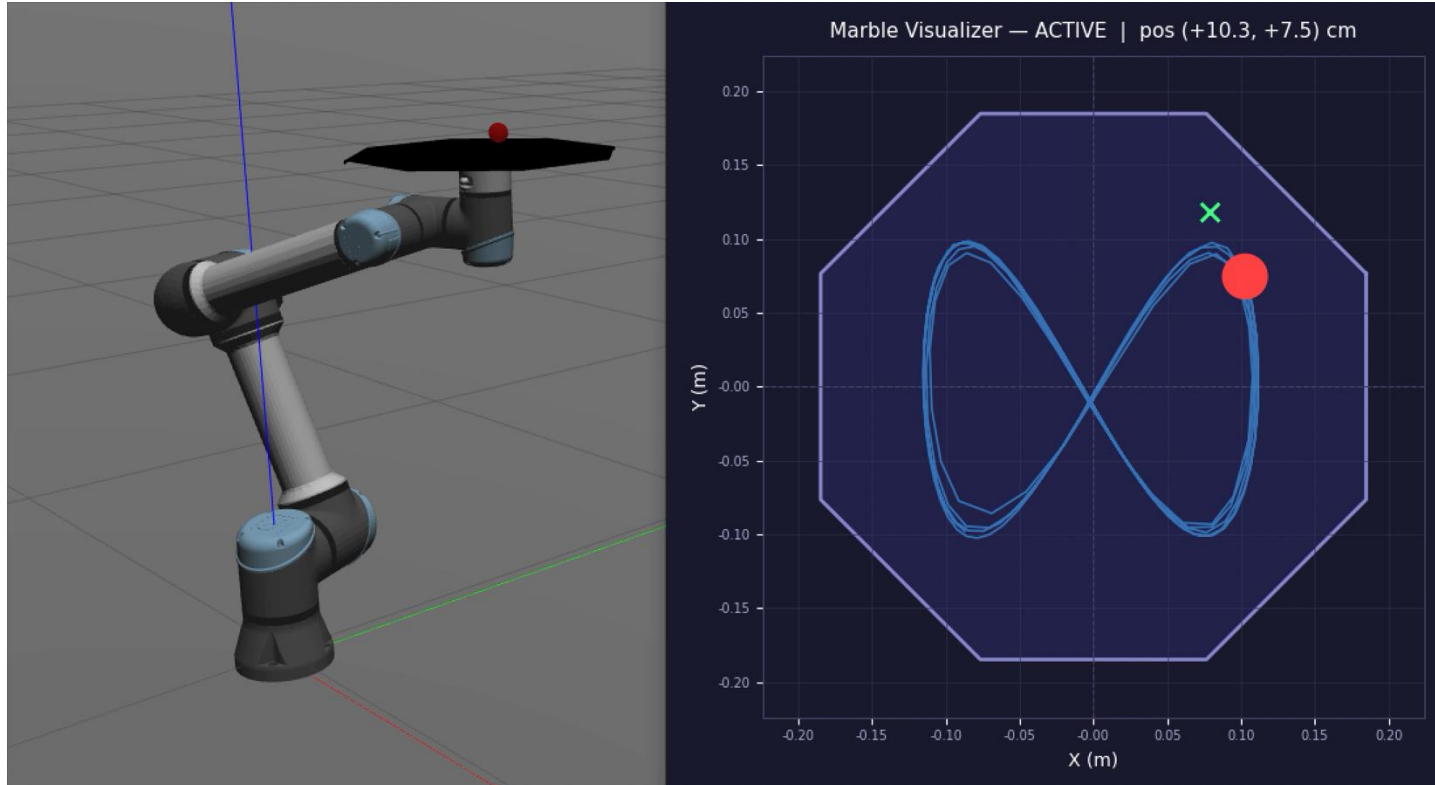
$$\mathbf{x} = [x \quad \dot{x} \quad y \quad \dot{y} \quad \alpha \quad \omega_\alpha \quad \beta \quad \omega_\beta]^\top$$



$$\mathbf{u} = [\omega_{\alpha,\text{cmd}} \quad \omega_{\beta,\text{cmd}}]^\top.$$

Demos /Results

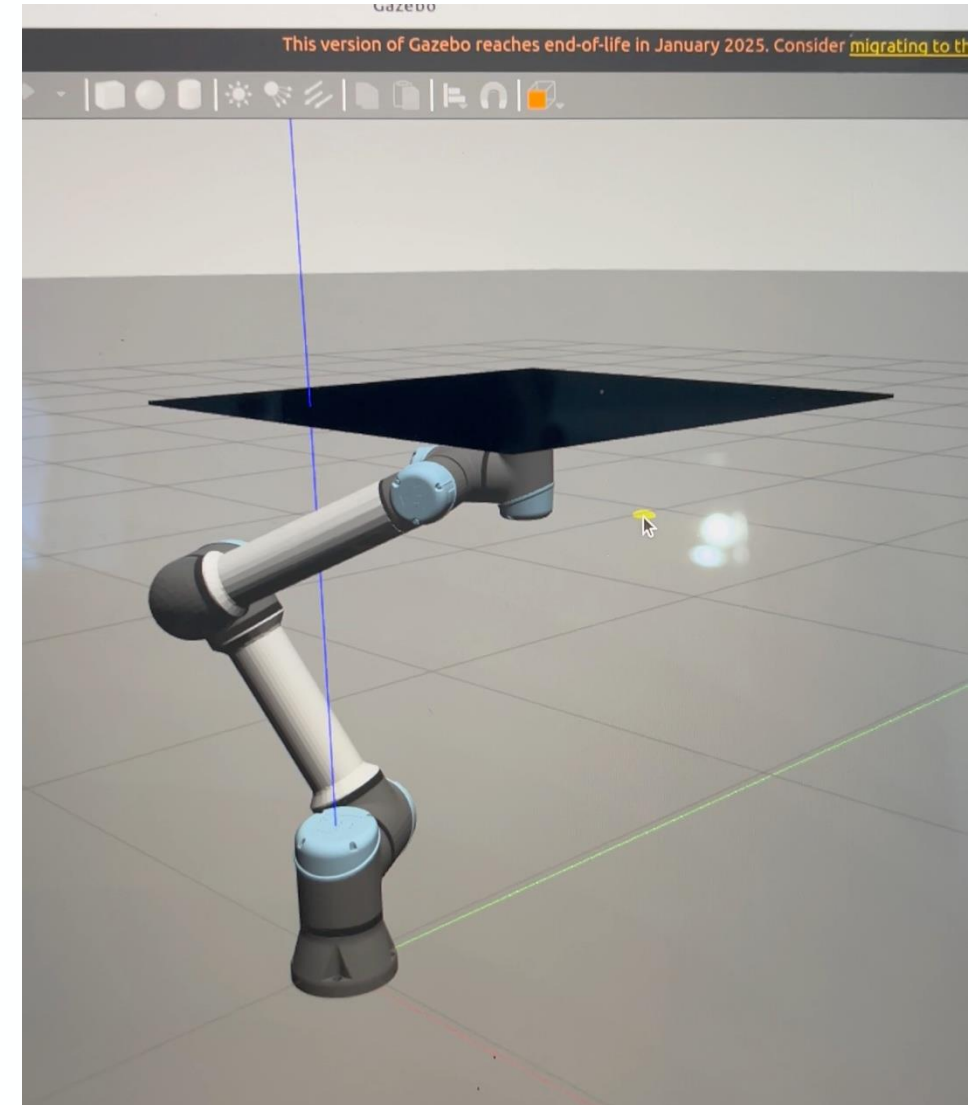
dLQR Controller



Methods

RL Residual Controller Overview

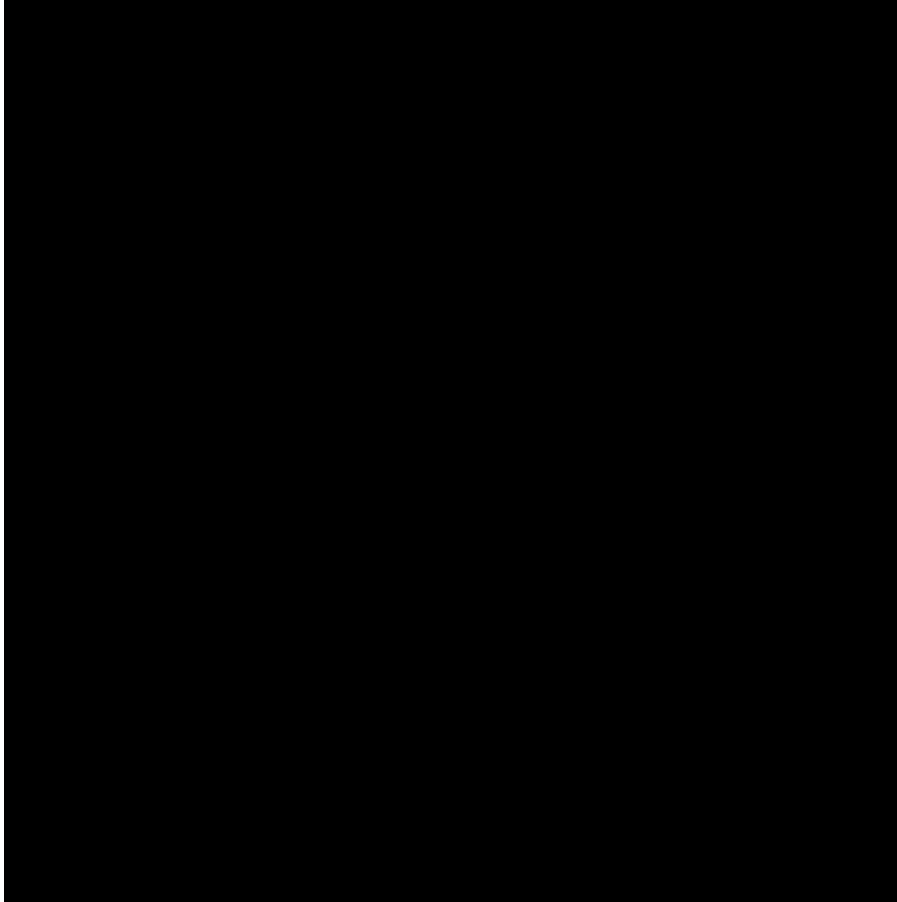
- Started with offline SAC.
- Moved to online SAC in Gazebo.
- Online SAC was not stable enough in the single real time Gazebo environment.
- Switched to TD3,
- Used dLQR as the baseline controller, used RL as a residual controller.
- Observation Vector
 - $o_t = [\text{ball position, ball velocity, plate angle, target position, plate angular velocity, previous position history}]$
- TD3 Reward
 - Position, Velocity Penalty, Smoothness penalty



Demos /Results

Hybrid Controller

- Residual RL ON



- Residual RL Off



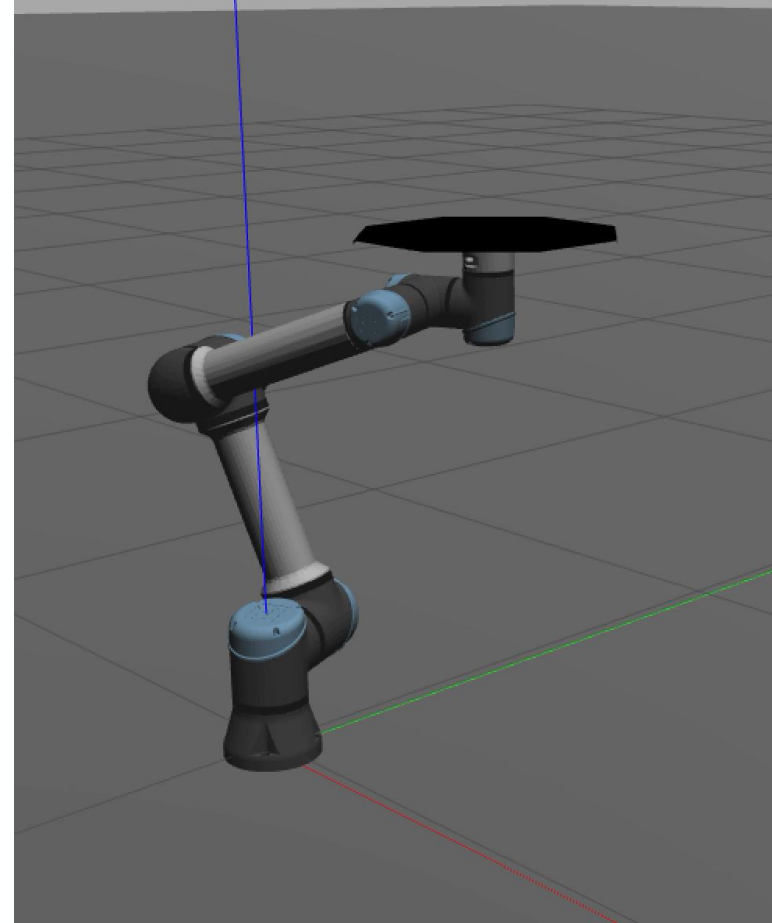
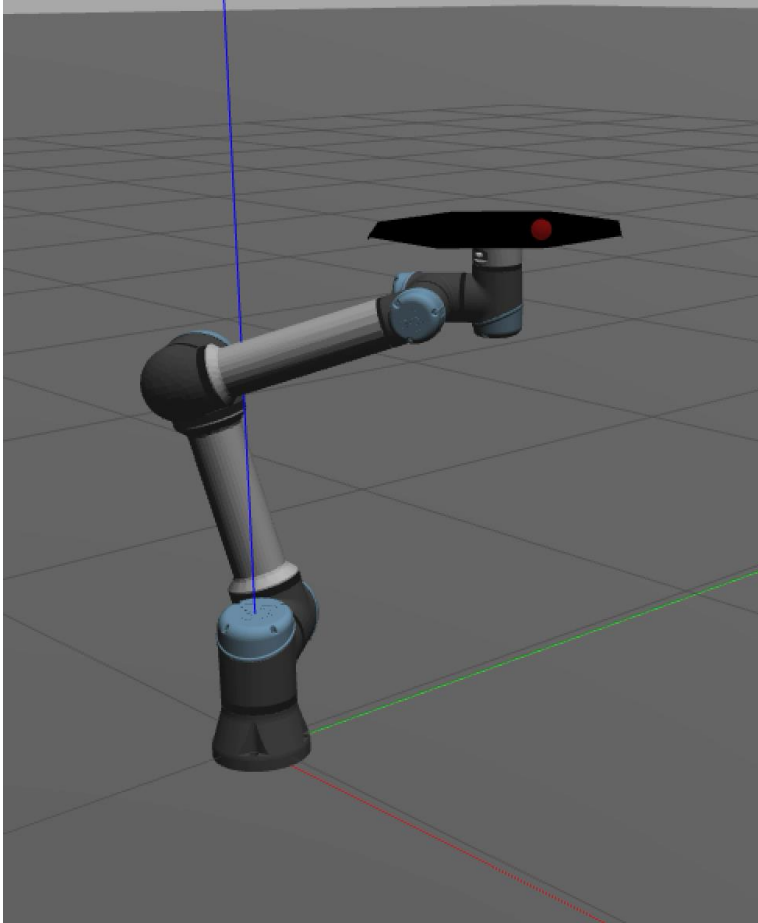
Methods

Improved Hybrid Controller

- Sharper position reward
- Symmetric reward
- Increased tilt penalty: $-0.2 \cdot (\alpha^2 + \beta^2)$ vs -0.1
- Removed x/y_dot velocity penalty
- Add Target position
- Reduced observation: 21-D vs 32-D → faster training
- TCP velocity window → learn acc forces
- Higher survival bonus: 0.1 vs 0.05 → encourages longer episodes

Demos /Results

Evaluation Lissajous & Jitter Walk



Demos /Results

Evaluation dLQR vs Residual

Test Scenario & Metric (200k Steps)	Pure LQR (Baseline)	Hybrid RL (TD3)	Delta / Improvement	Performance Verdict
Scenario 1: Dynamic Disturbance (Lissajous TCP)				
Average Total Reward	+711.99	+690.81	-3.0%	LQR (-)
RMSE (m)	0.1920	0.1898	+1.1%	RL (+)
Max Peak Error (m)	0.3858	0.4098	-6.2%	LQR (-)
Episode Win Rate (rew)	40% (4/10)	60% (6/10)	+20.0%	RL (+)
Scenario 2: Extreme Stress Test (Lissajous TCP, 0.18m Edge Spawn + Jitter)				
Average Total Reward	+823.68	+778.91	-5.4%	LQR (-)
RMSE (m)	0.2587	0.2556	+1.2%	RL (+)
Max Peak Error (m)	0.6923	0.6379	+7.9%	RL (+)
Episode Win Rate (rew)	70% (7/10)	30% (3/10)	-40.0%	LQR (-)

What remains to be done

- Improve Hybrid Controller performance through:
 - Reward shaping
 - Hyperparameter tuning
 - Bigger Training Data/Sample
 - More Cases generalizable
- Goal
 - Performance improvements over all scenarios
 - Specially reducing the max error and RMSE

Future Work

- Deploy the hybrid controller on the real UR5e robot
- Integrate camera-based marble tracking for real-world sensing
 - Object tracking or Color segmentation tracking
- Further tune the EKF noise parameters for noisy hardware measurements
- Extend sim-to-real transfer and test robustness under real disturbances

Team member contributions

	Project Contribution	Team Member
1.	Set up Gazebo Server	Kevin
2.	Integration of Generic Ur description	Phillipp + Kevin
3.	Integration of Marble + Plate	Phillipp
4.	Integration/config of low level PD Joint controller	Phillipp
5.	Integration of MoveIt-Servo (low latency vel-command) ROS2 node	Phillipp
6.	System state estimation Jacobian vs. derivative	Phillipp
7.	Mathematical System Modeling Marbel + Plate + Robot	Phillipp
8.	System discreditation and Performance index design	Phillipp
9.	Design of Optimal dLQR-Controller ROS2 node	Phillipp
10.	Vel-command mux TCP Rot(LQR) + Trans-vel à TCP_vel(Low-level)	Kevin
11.	Extended Kalman filter for State Estimation	Phillipp
12.	Integration of Online and Offline SAC-RL residual controller	Kevin
13.	Integration of TD3 RL residual controller	Kevin
14.	CMAES Validation	Kevin
15.	RL residual controller training and Optimization	Kevin + Phillipp
16.	Monitoring & Evaluation Tools (plotter, compare controller tools)	Phillipp

Thank You

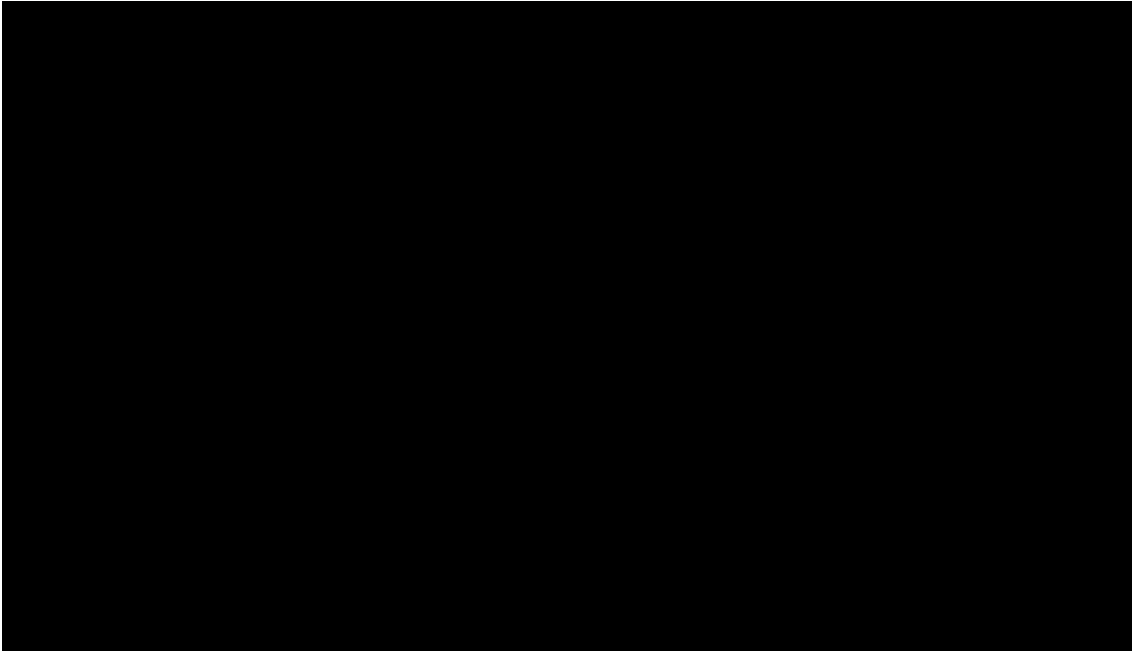


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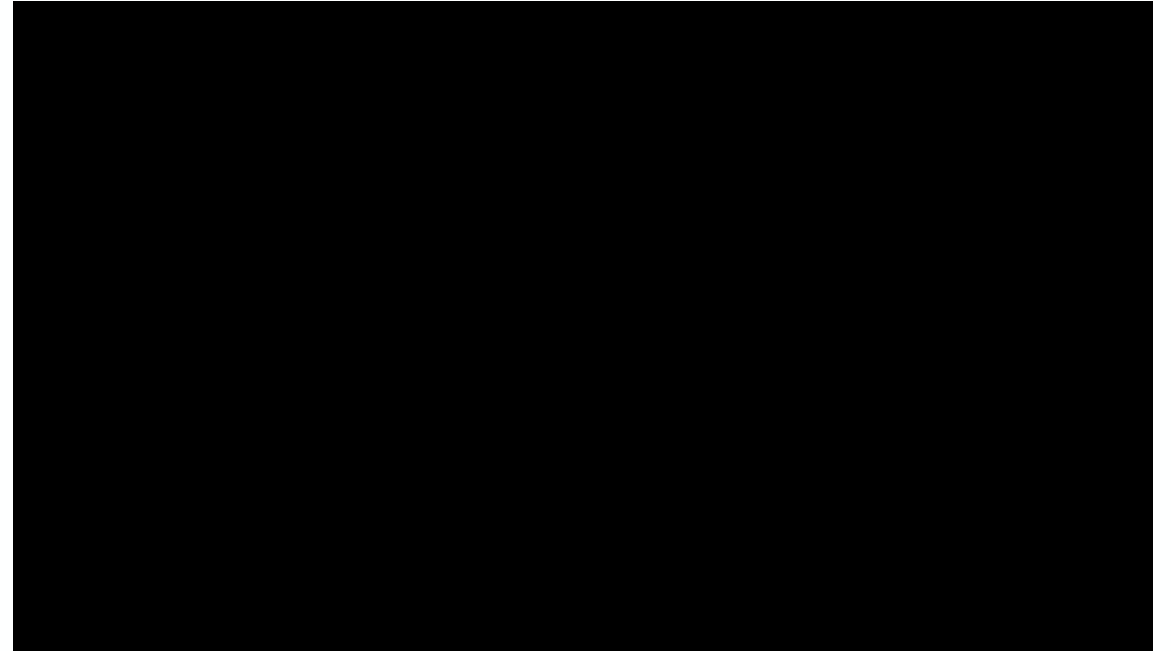
Demos /Results

Hybrid Controller

- Residual RL ON



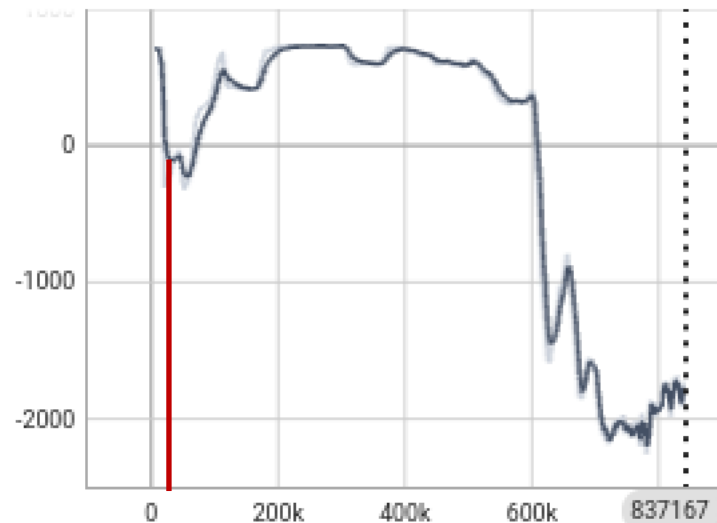
- Residual RL Off



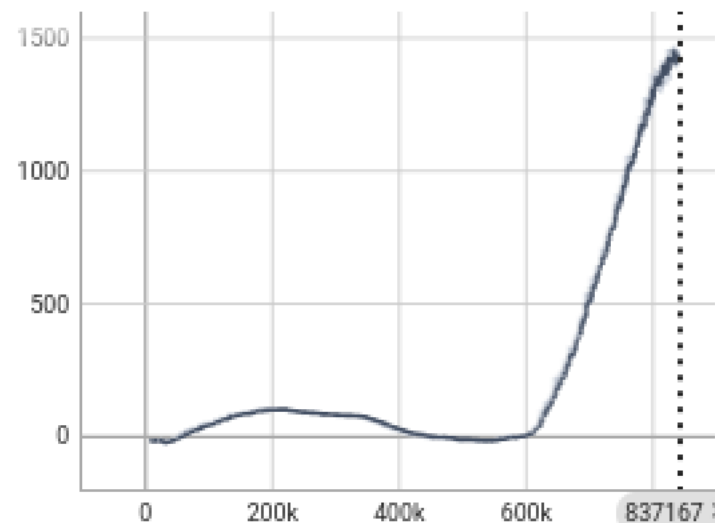
Demos /Results

Hybrid Controller

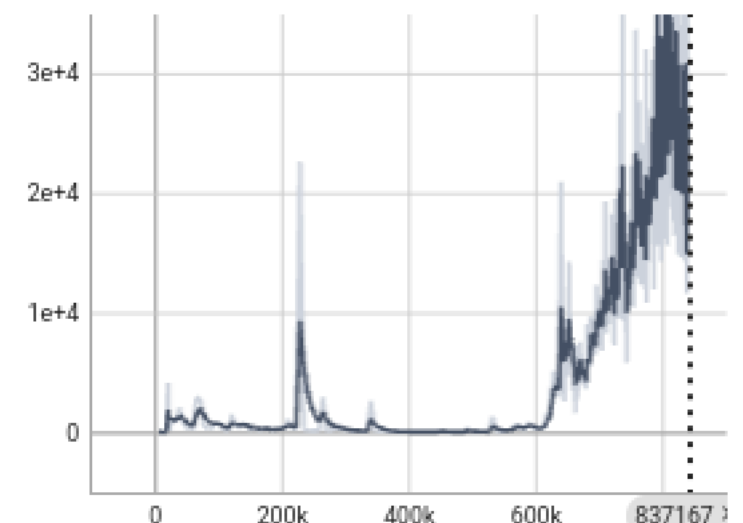
rollout/ep_rew_mean



train/actor_loss



train/critic_loss

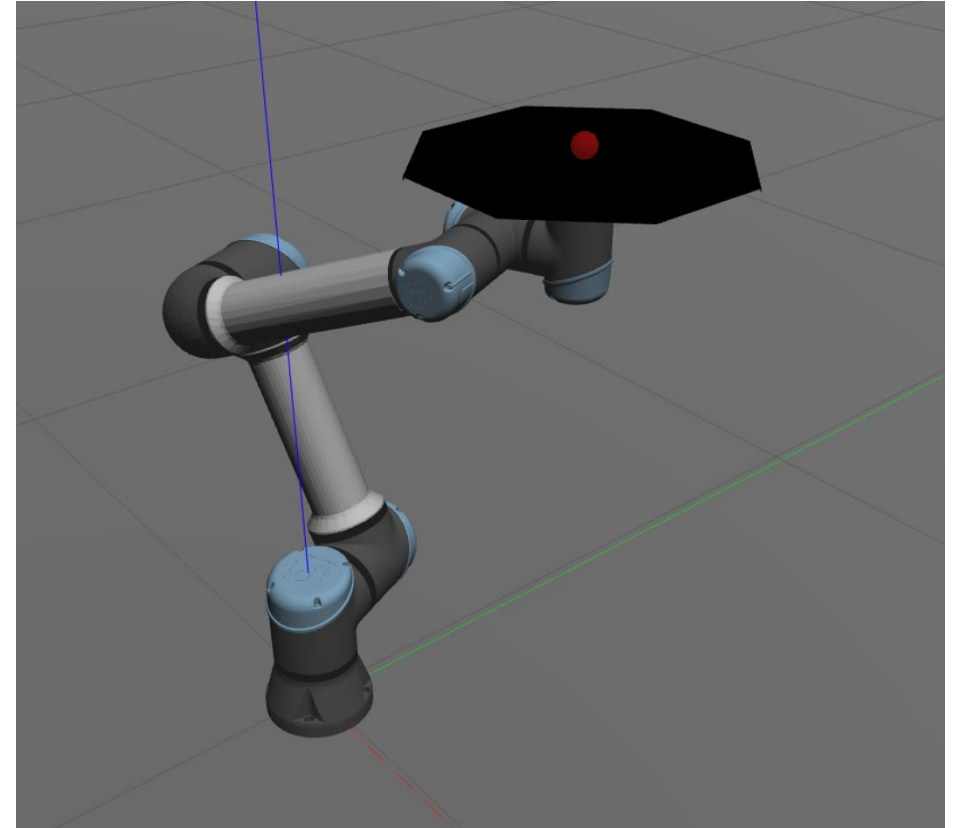


Methods

- Developed a hybrid control architecture for 6-DOF ball-and-plate stabilization using a UR5e robotic arm in Gazebo.
- Used a discrete-time LQR controller (30 Hz) as the baseline optimal controller for ball centering.
- Modeled the system with an 8-state ball-and-plate representation including ball position, velocity, plate angles, and angular rates.
- Applied an Extended Kalman Filter (EKF) to estimate the full system state from measurements.
- Trained a residual reinforcement learning policy (TD3/SAC) to generate corrective actions on top of the LQR output.
- Used a 21-dimensional observation space including system state, setpoints, LQR commands, and TCP motion data.
- Combined both controllers as final control = LQR + residual RL, improving disturbance rejection and tracking performance.
- Evaluated the method under dynamic disturbances such as jitter walk and Lissajous TCP motion.

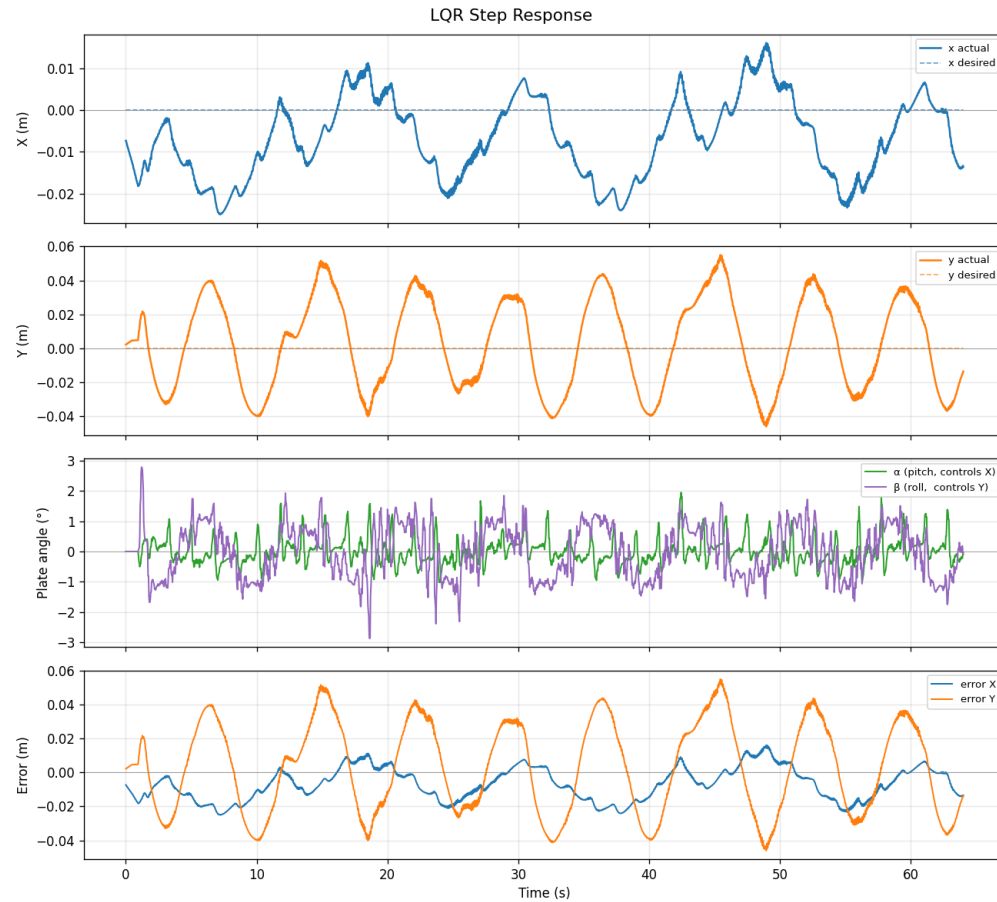
Introduction

- This project focuses on stabilizing a ball and plate system mounted on a 6-DOF robotic arm, a task made challenging by nonlinear dynamics, coupling effects and external disturbances. A hybrid control strategy is used, where LQR provides reliable baseline stabilization and residual reinforcement learning adds corrective actions to improve performance when disturbances or model mismatch occur. This combines the stability of classical optimal control with the adaptability of learning-based control for more robust ball balancing.



Demos /Results

■ Residual RL ON



■ Residual RL Off

